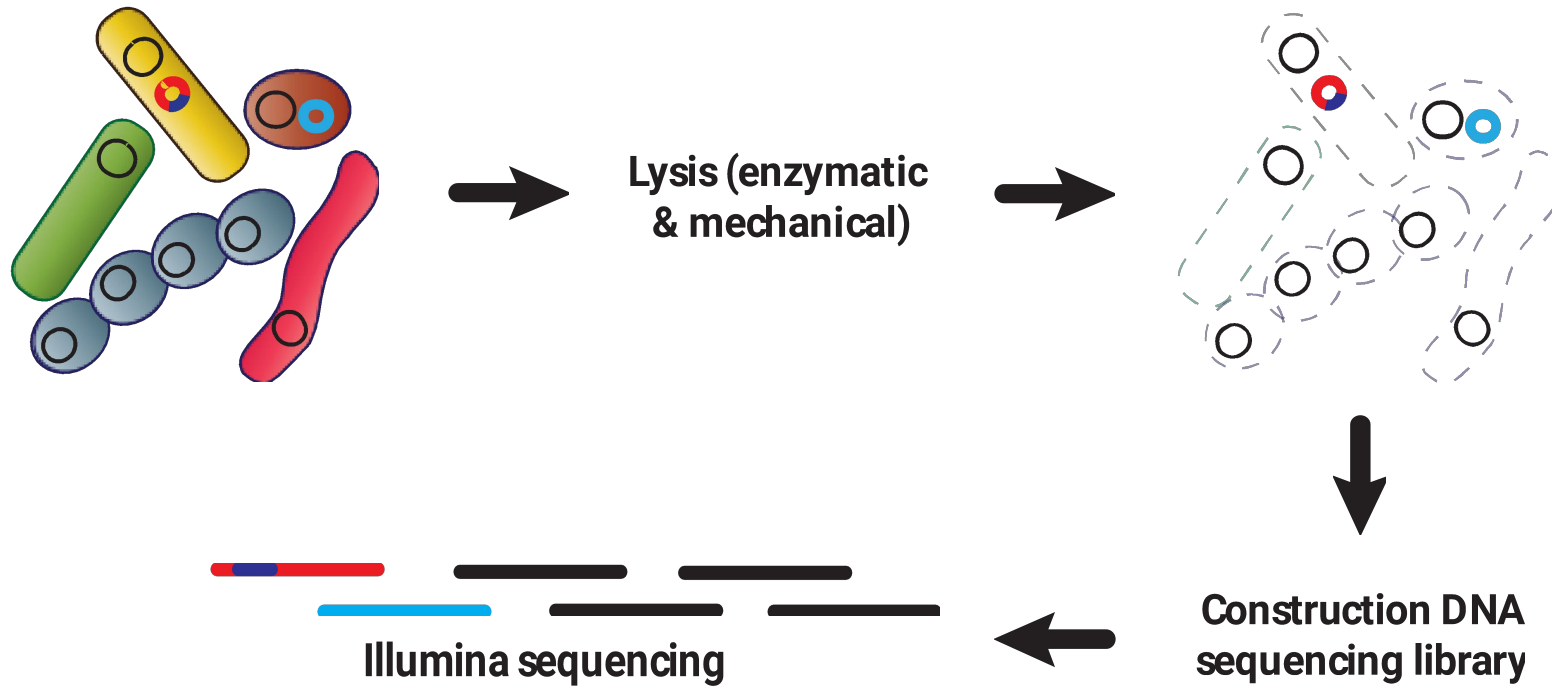


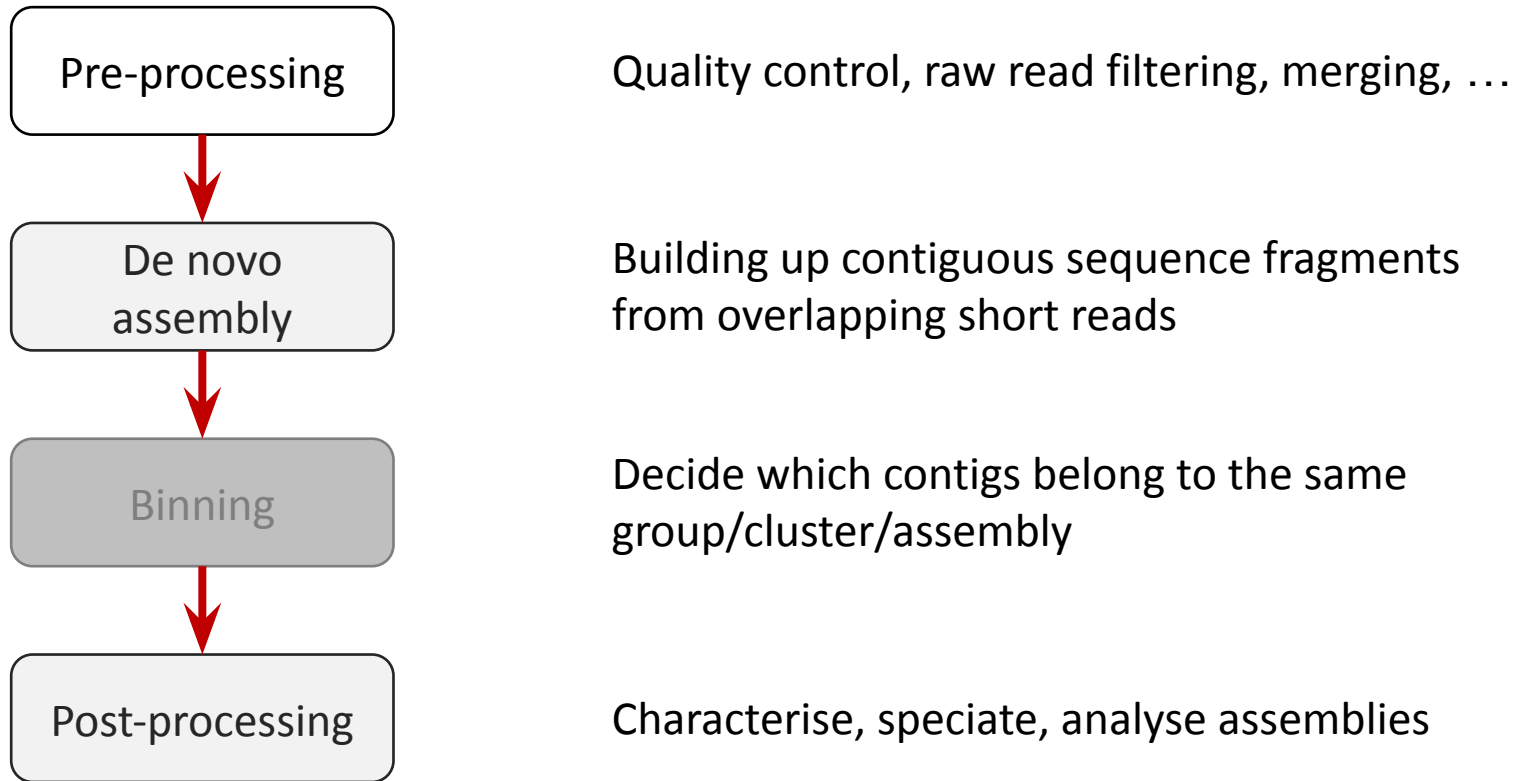
Shotgun *de novo* metagenomics

Lajos Kalmar, MRC Toxicology Unit, University of Cambridge

Basic laboratory workflow



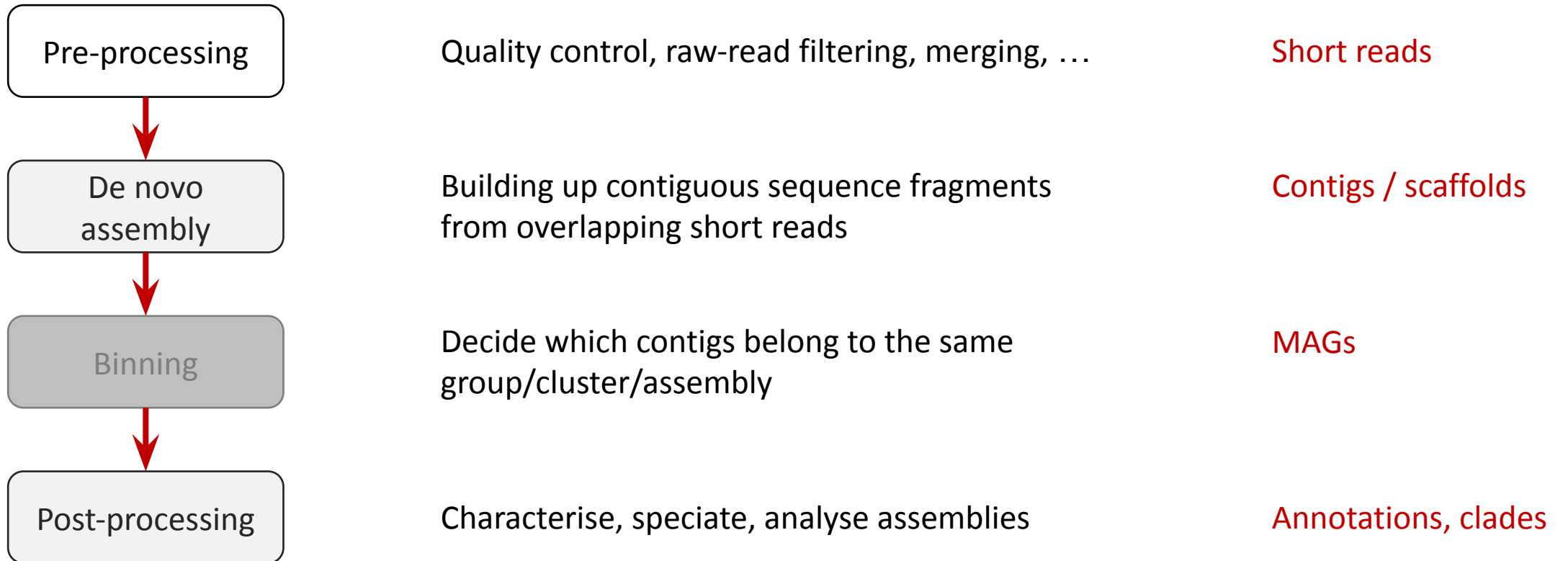
Basic bioinformatics workflow



List of definitions

- Short reads: sequence reads from NGS
- Contigs: continuous DNA sequence (/[ACGT]/)
- Scaffolds: contigs glued together in their most probable orientation and distance (CONTIG1NNNNNNNCONTIG2NNNNNNNCONTIG3)
- Assembly: Collection of contigs / scaffolds in a multi-fasta file
- MAGs: Metagenome assembled genomes
- Clade: a group of assemblies that are considered “similar” by e. g. average genetic distance

Basic bioinformatics workflow



Ideal world 1 – reference-based re-alignment

- We know all bacterial genomes that are potentially present in the analysed community and have good references for each of them
- Align all short reads to reference sequences (bowtie2, bwa)
- Extract unique alignment information (presence / absence)
- Extract coverage information (abundance)
- Extract SNP and short indel information (genetic variance)
- Extract structural variations (genomic rearrangements)
- Gold standard whole-genome re-alignment pipelines

Ideal world 1 – reference based re-alignment (potential real world examples)

- You are testing different probiotics in the relation of the presence / abundance of an unwanted bacterial species in the human colon (colonisation resistance)
- Database build-up: the unwanted bug, all probiotic species, a few well known commensals from human colon microbiota
- Take multiple samples from patients during the probiotic treatment and check the good (commensal & probiotic) / bad (unwanted) bacteria ratio
- Extreme but clinically relevant example: FMT (faecal microbiota transplant) in *Clostridium difficile* infection

Ideal world 2 – de novo assembly of whole genomes

Step 1 – finding overlapping short reads with multiple kmers



Step 2 – Extending sequence



Repeat until
circularisation

Start over with a
new short read

Reality – Intra and inter species repeats

- Homologous and multi copy genes
- Transposons, phage sequences, other mobile genetic elements



B. caeruleus



B. viridis

Assembling contigs with repeat sequences



B. caeruleus

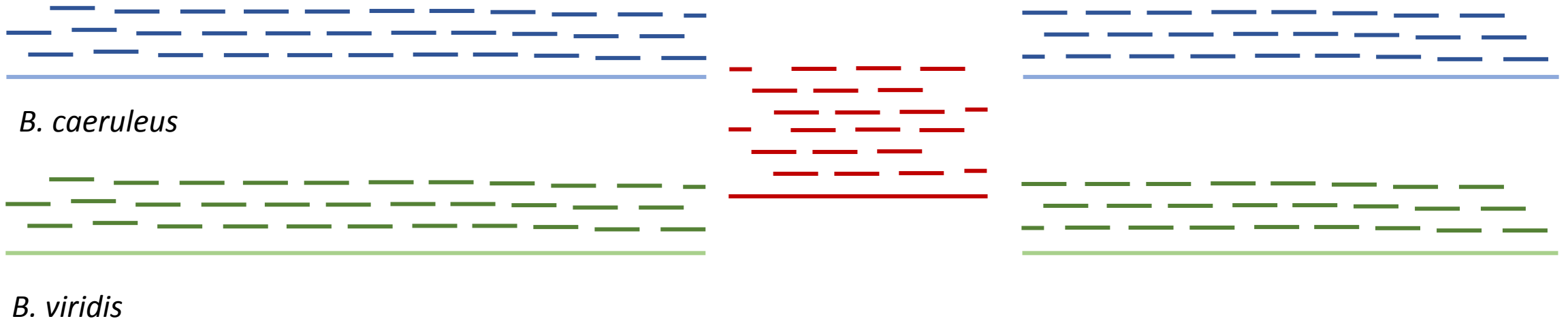


B. viridis

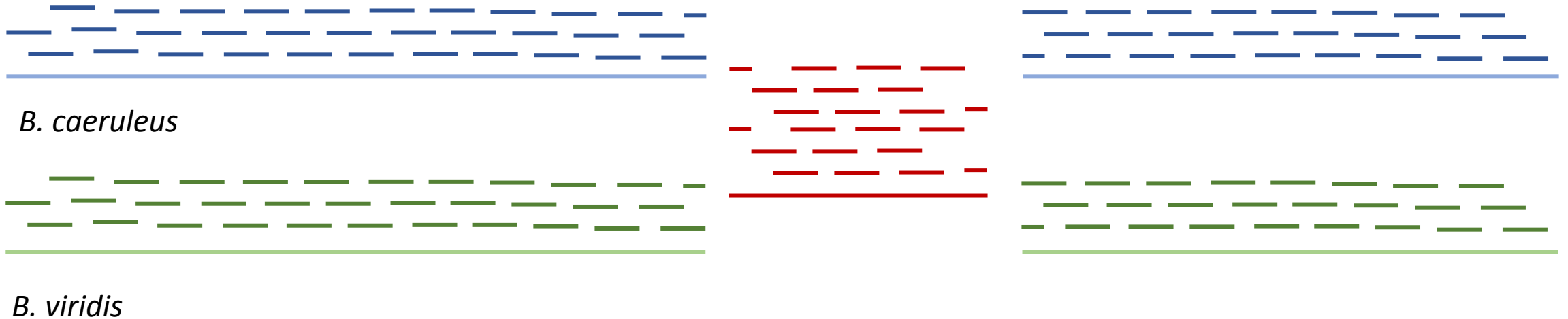
BC_GenomeFragment_1 → BC_GenomeFragment_2 → BC_GenomeFragment_3

BV_GenomeFragment_1 → BV_GenomeFragment_2 → BV_GenomeFragment_3

Assembling contigs with repeat sequences

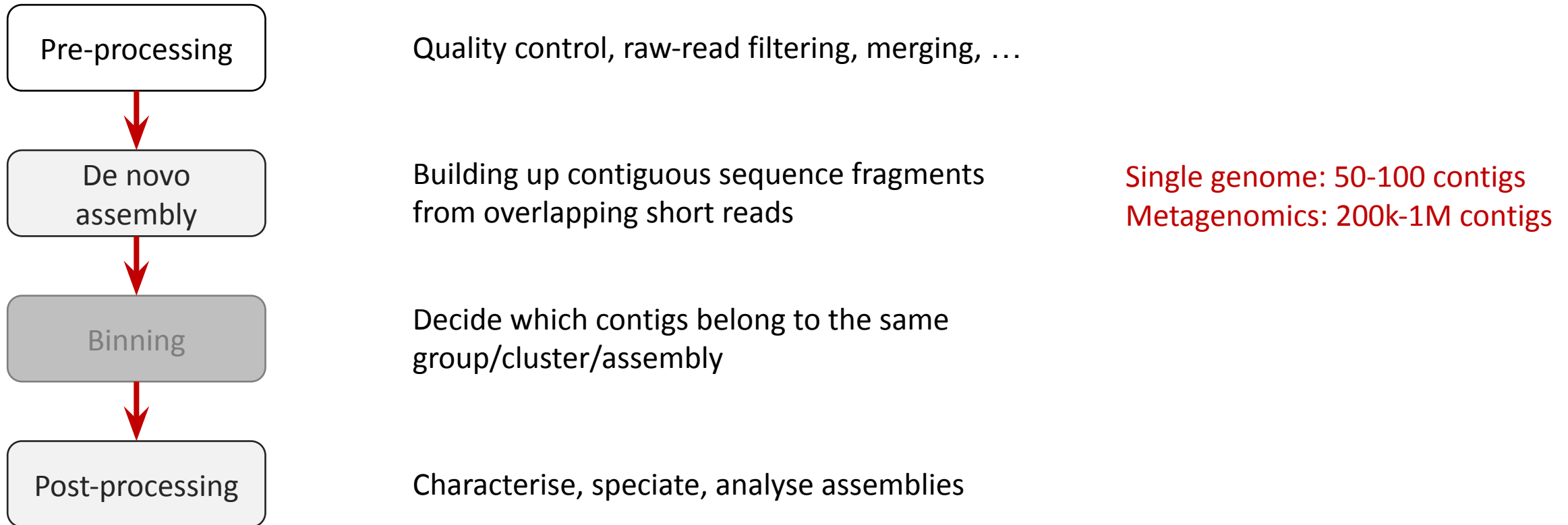


Assembling contigs with repeat sequences



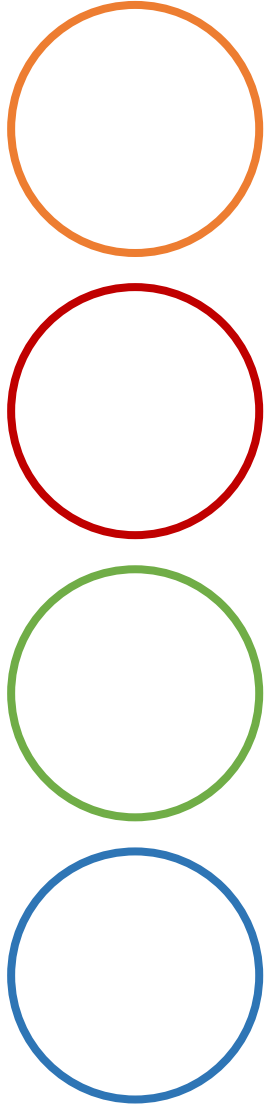
BC_GenomeFragment_1coverage X
BC_GenomeFragment_3coverage X
REP_GenomeFragment coverage X+Y
BV_GenomeFragment_1coverage Y
BV_GenomeFragment_3coverage Y

Basic bioinformatics workflow

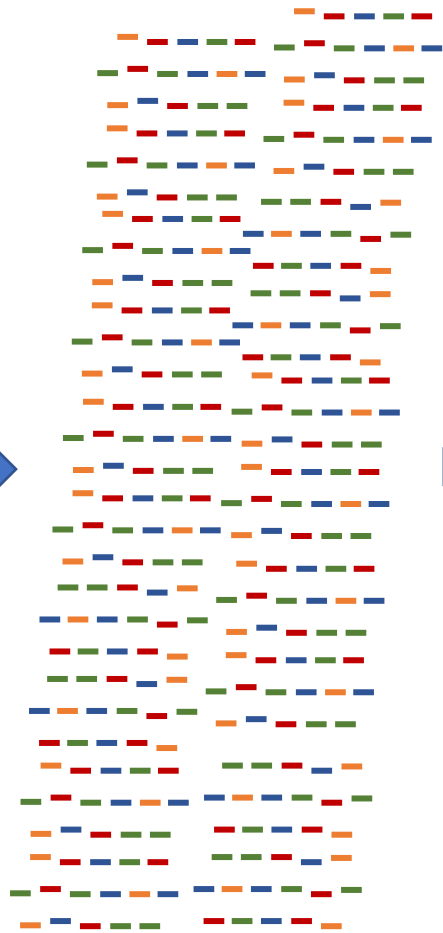


Binning – finishing the puzzle

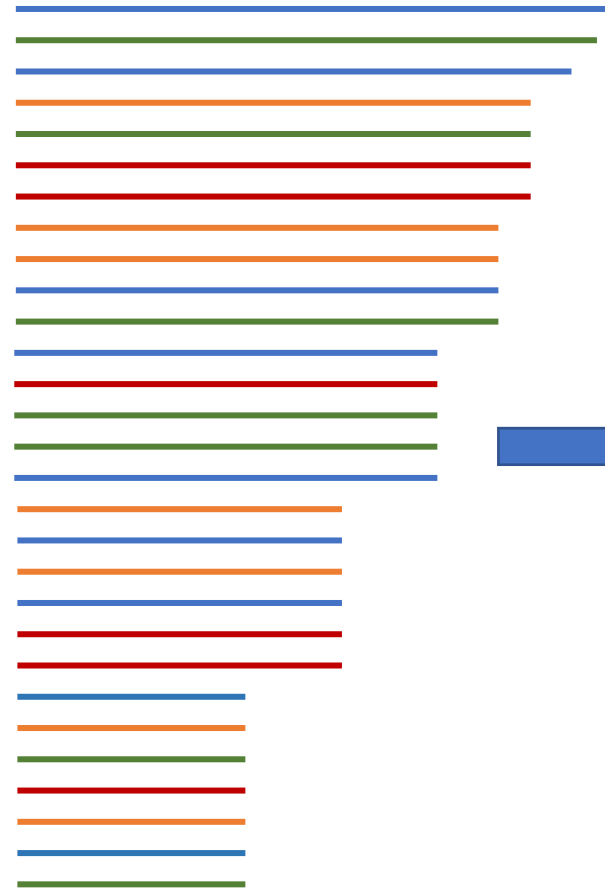
Original genomes



Shotgun sequencing



De-novo assembly



Binned final assemblies



Metagenomics binning approaches

Supervised

Using known reference sequences to group contigs together

- Fast, reliable, simple
- High dependency on prior knowledge
- Can introduce species bias
- Masks segmental rearrangements
- Good for well known microbiomes
- Unreliable for less known microbiomes

Unsupervised

Using genomic and statistical properties to group contigs together

- Slow, complex, universal
- No prior knowledge needed
- Works better with multiple samples
- Problem with resolution on lower taxonomic levels
- Good for less known microbiomes
- Not as specific for known microbiomes as supervised

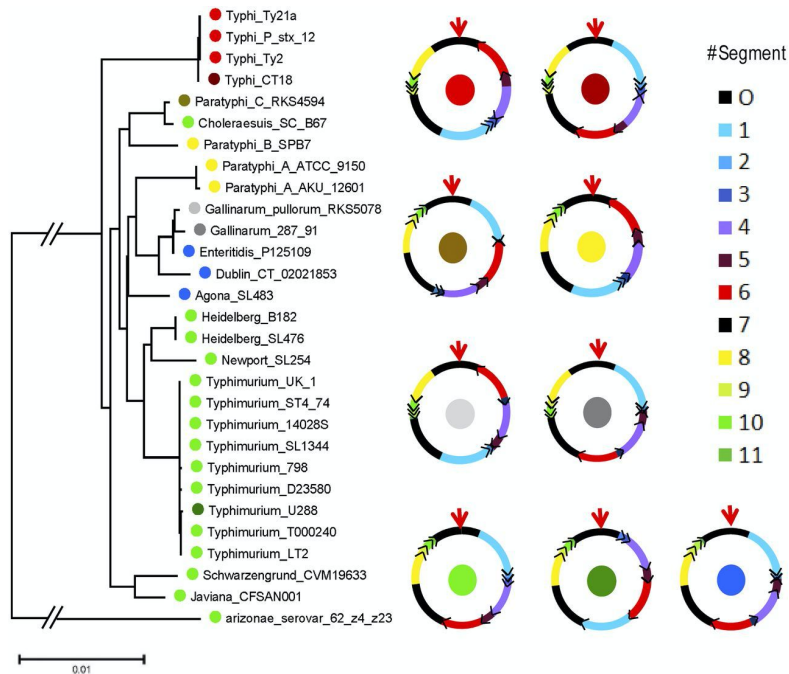
Long read sequencing

Using long-read sequencing in combination with shotgun sequencing

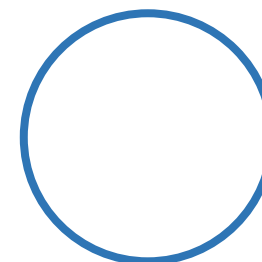
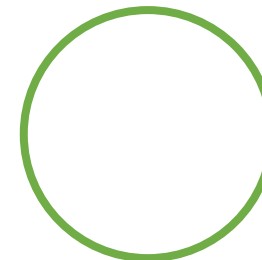
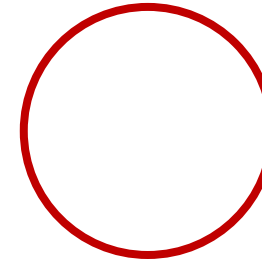
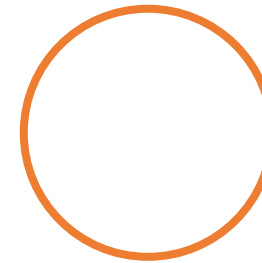
- Slow, complex, universal
- No prior knowledge needed
- Can detect segmental rearrangements
- Expensive
- Lack of hybrid software applications, pipelines
- Method specific issues with accuracy
- The future ultimate solution?

Supervised binning

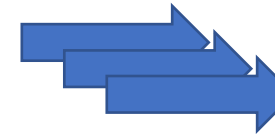
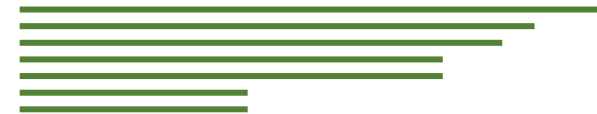
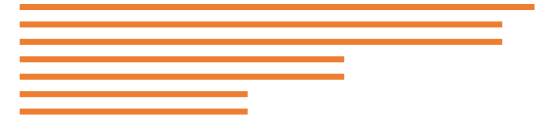
- High chance of not assembling previously unknown species
- Relative abundance biased
- Low chance to detect segmental rearrangement



Original genomes

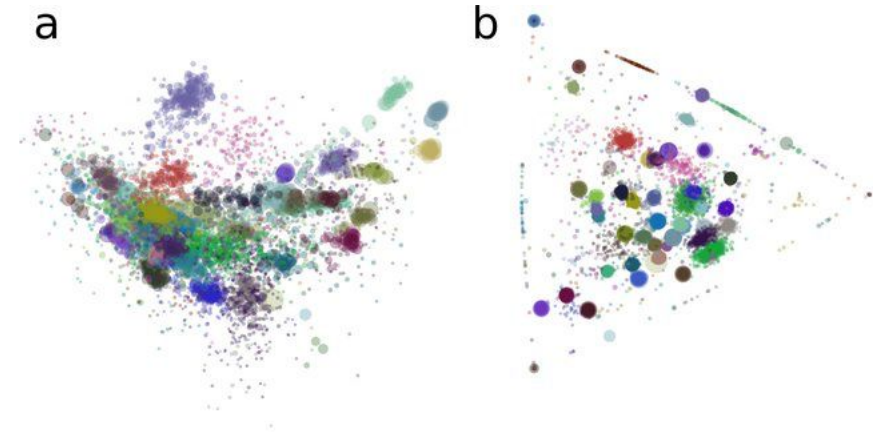


Binned final assemblies



Unsupervised binning

- Original idea: GC content vs. coverage
- Recent methods: PCA analysis (multiple coverages, GC, tetranucleotide kmer frequencies)
- Best practice:
 - Having multiple samples from the same microbiome (spatial or temporal difference)
 - Do the de novo assembly using a few samples
 - Re-align contigs with short reads from all samples and calculate coverages
 - Do the unsupervised binning with multiple sample coverage data
- Poor differentiation power for low abundance species, especially if there are multiple close species
- Novel genetic elements? Horizontal gene transfer?
- Multi-copy genes!



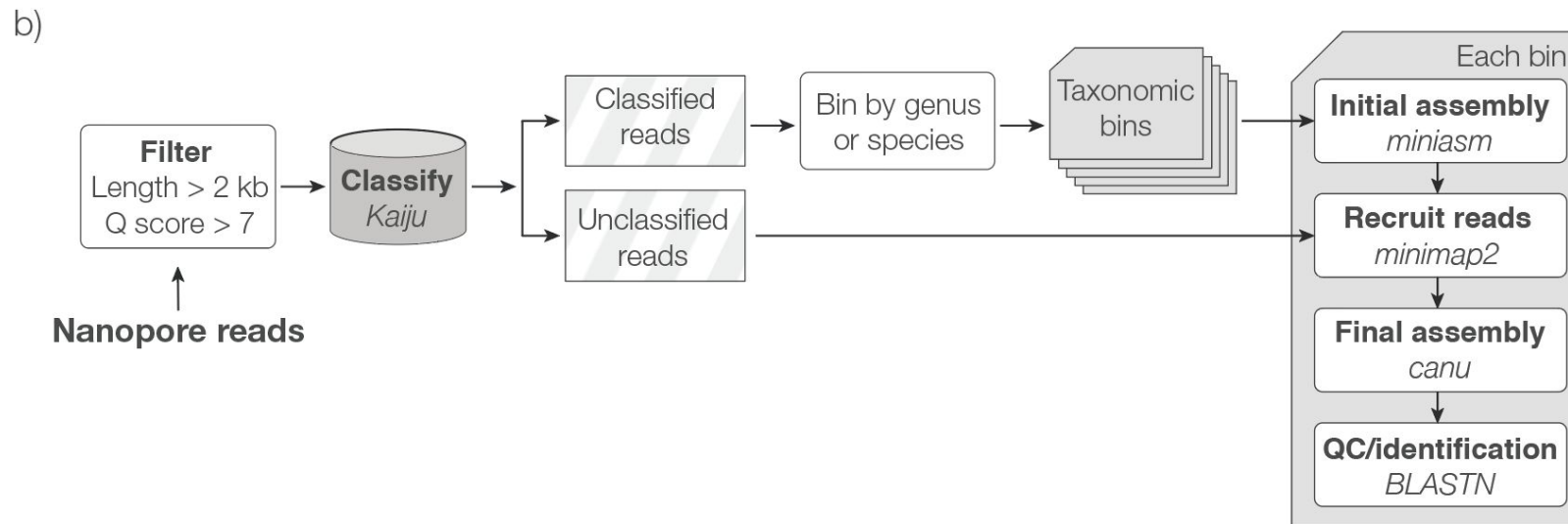
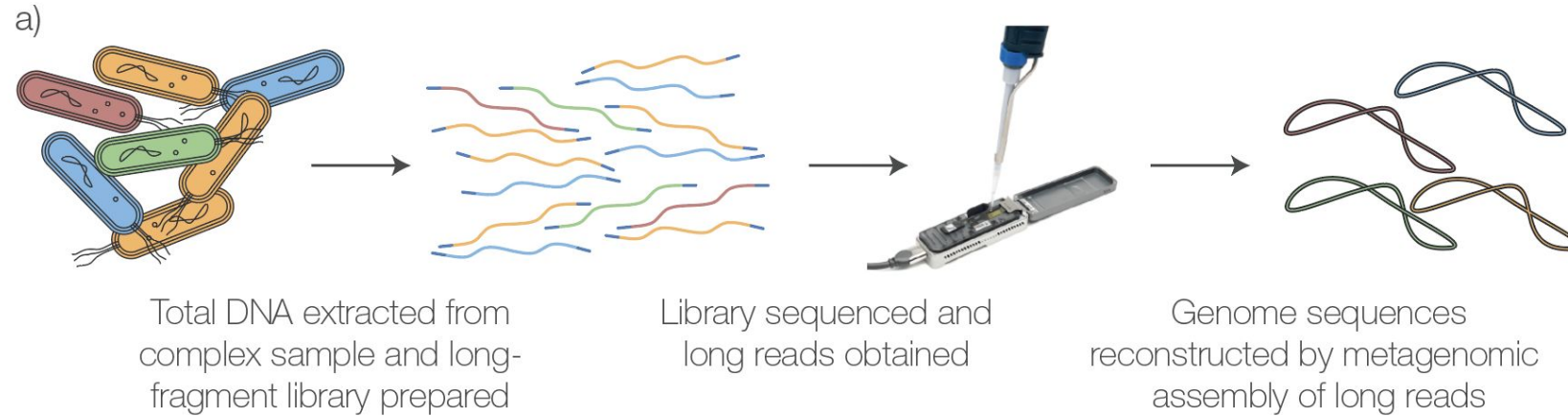
Using long reads

- Long read sequencing
 - PacBio: good quality, very expensive
 - Oxford nanopore: quality not good enough alone, (less) expensive
- Special sample handling and DNA preparation
- Should be used in combination with short read sequencing
- Lack of good, user friendly hybrid pipelines
- More bioinformatics skill required

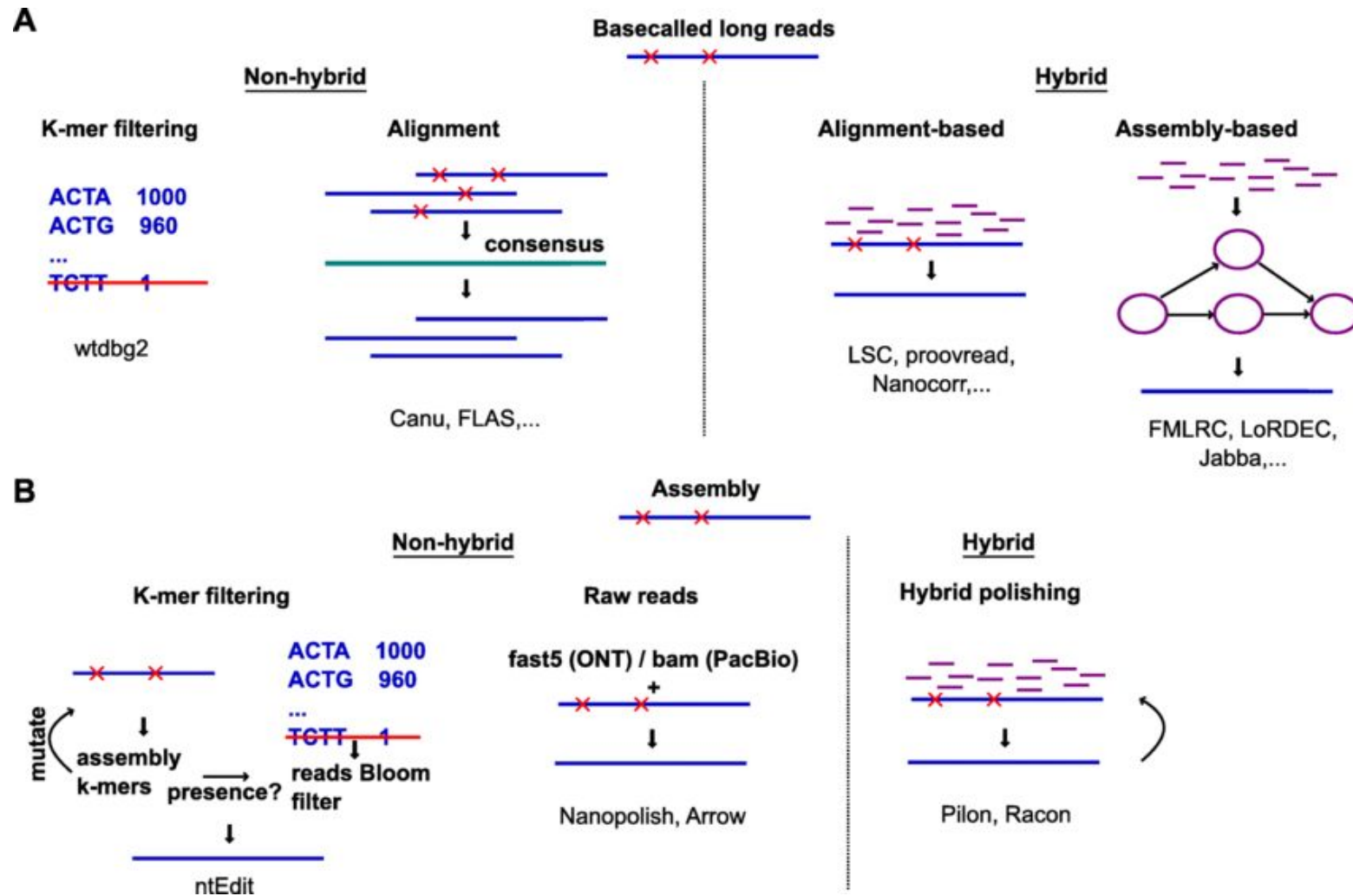
Long read sampling & preprocessing

- Sample handling, lysis and shearing need to be done with extra care to keep high molecular weight DNA extraction possible
- Large samples with high DNA yield can be run through size selection
 - Might effect quantitative results
- 'Quality scores' unavailable or have very different meaning to the ones we know from short reads
- Preprocessing tools/steps depend on technology (Guppy, proprietary PacBio tools)

Using long reads for assembly



Polishing (or Anchoring)



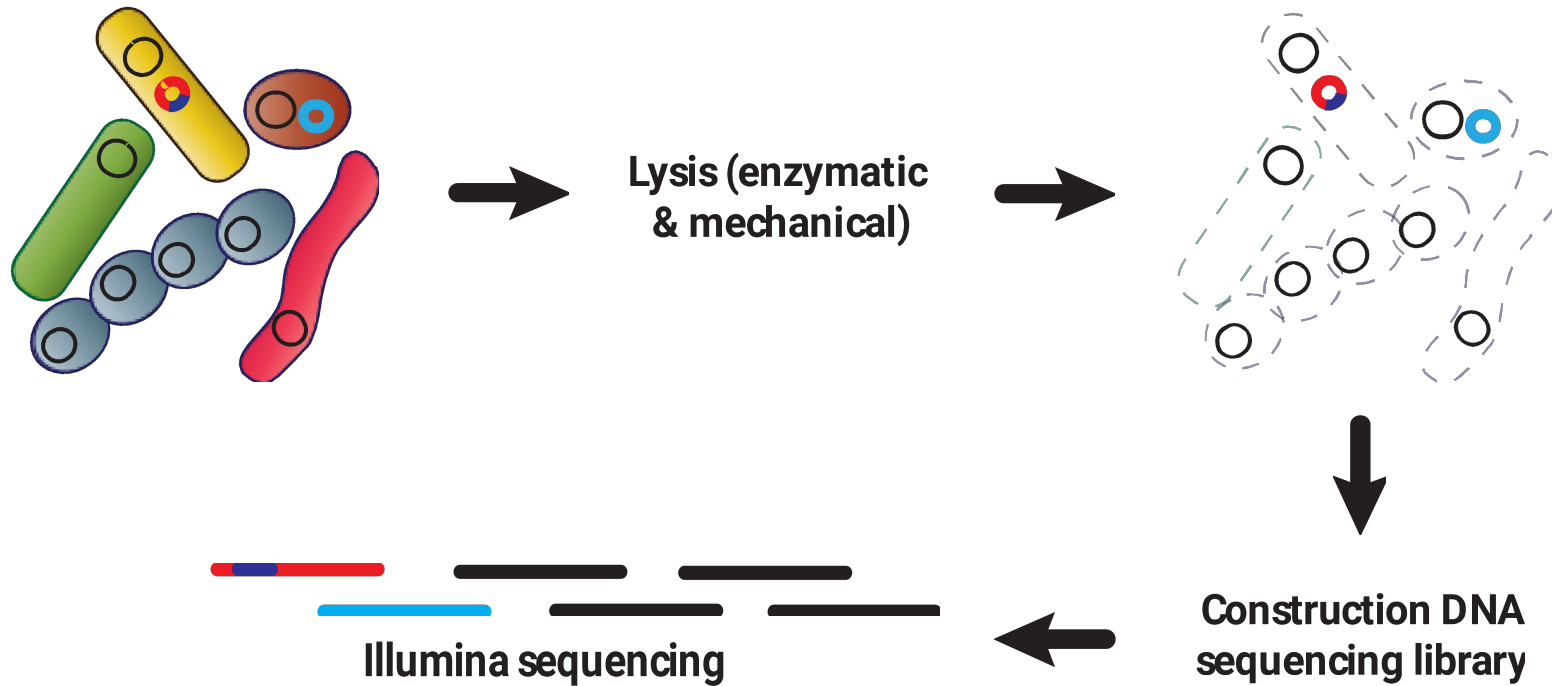
Long read metagenomics tools

- Preprocessing: Guppy
- Assembly: unicycler, flye, canu
- Alignment: DIAMOND, BWA
- Classification: Kaiju, MEGAN, bugSeq
- Polishing: Arrow, Pilon

Issues:

- High hardware requirements (RAM!)
- Pipelines can be complex
- Need for polishing or other post-processing steps (frameshift correction)

Mobile genetic elements?



Hi-C metagenomics

Antimicrobial resistance - AMR



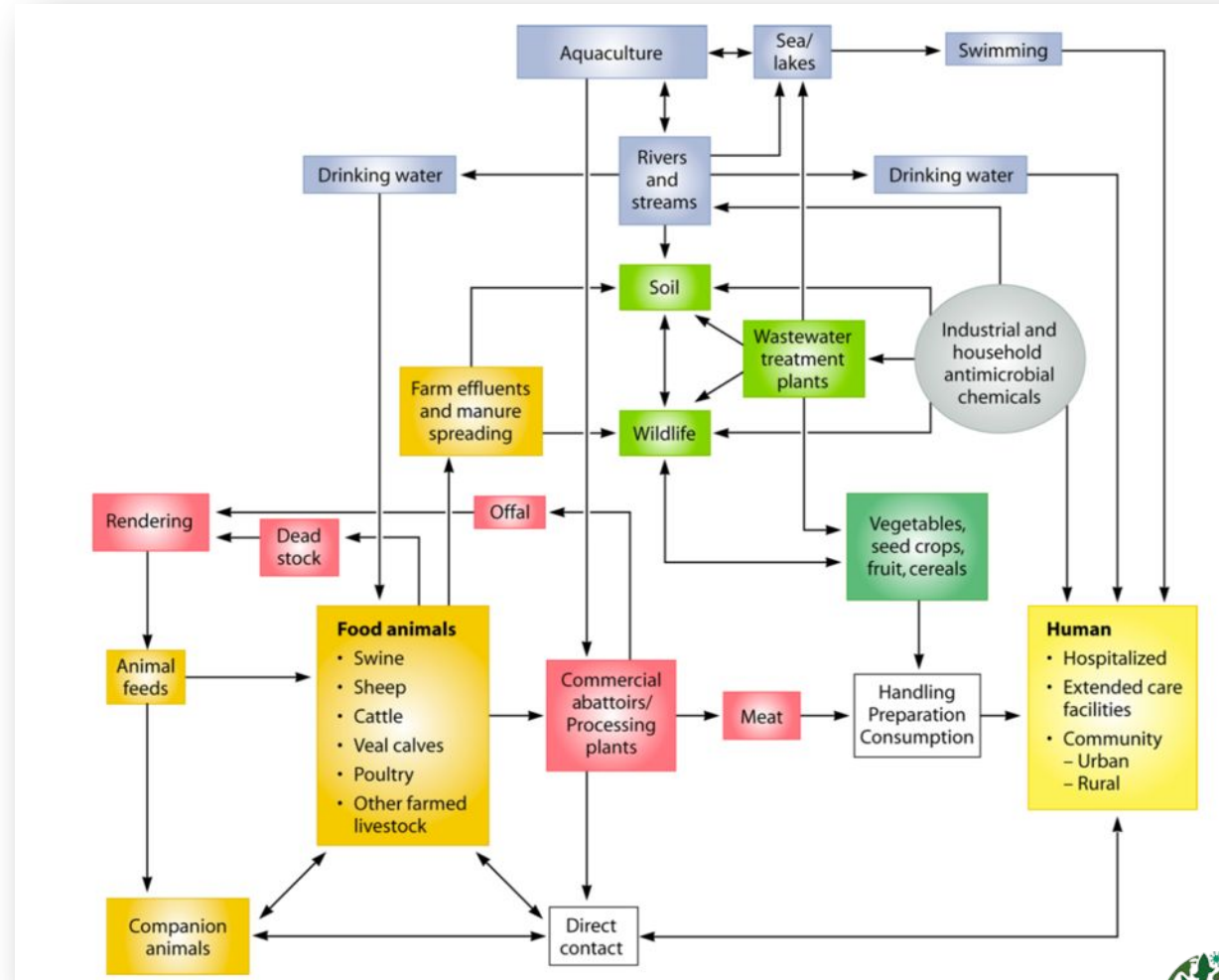
- One of the 10 threats to global health
- Resistance emerging in years, spreading very fast
- Multifactorial problem mostly in developing countries
- Discover new antimicrobial compounds
- Understand the dynamics of resistance transmission
- Prudent use of recently available antibiotics:
 - **Appropriate Drug / Dosage / Duration**

Antimicrobial resistance - AMR

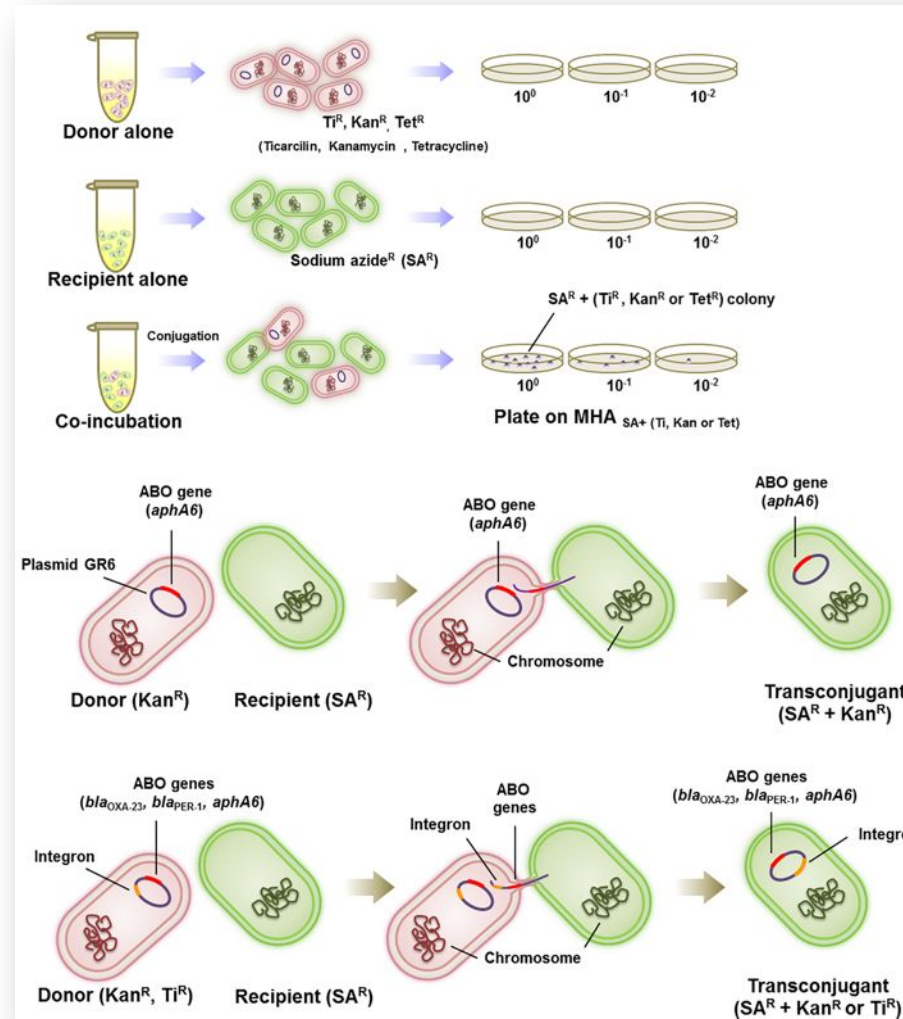


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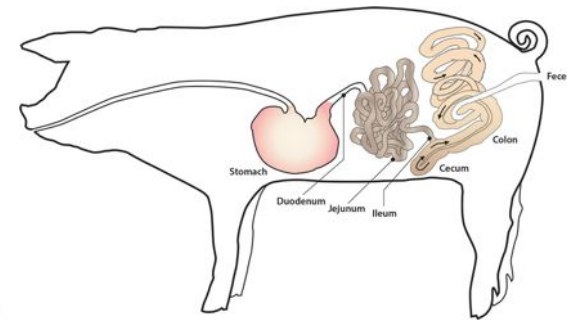
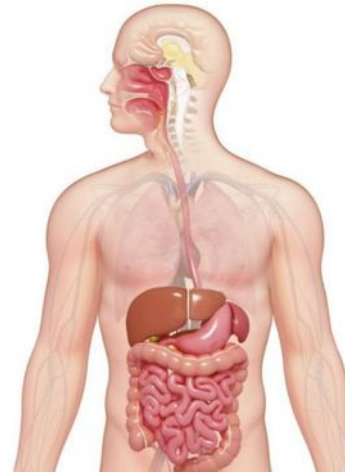
Dissemination of antibiotics and antibiotic resistance



This is not enough anymore



Antibiotics selection pressure on complex microbial communities

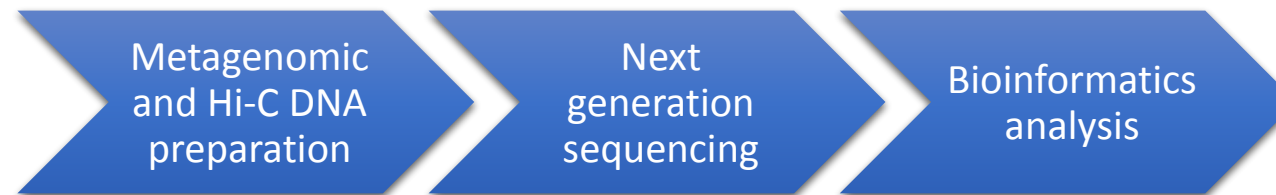


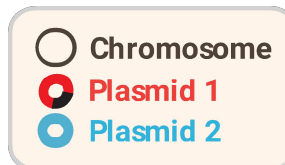
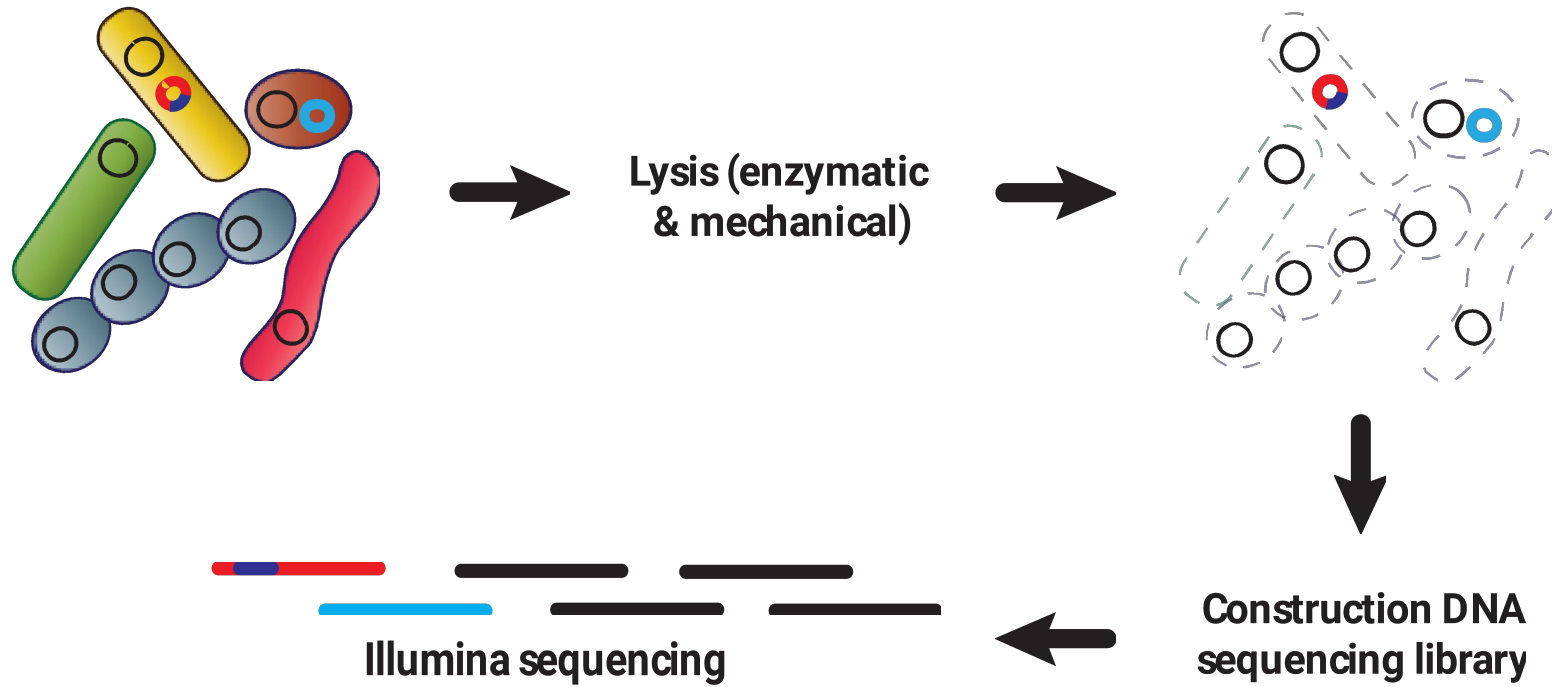
Identify species in complex microbial communities

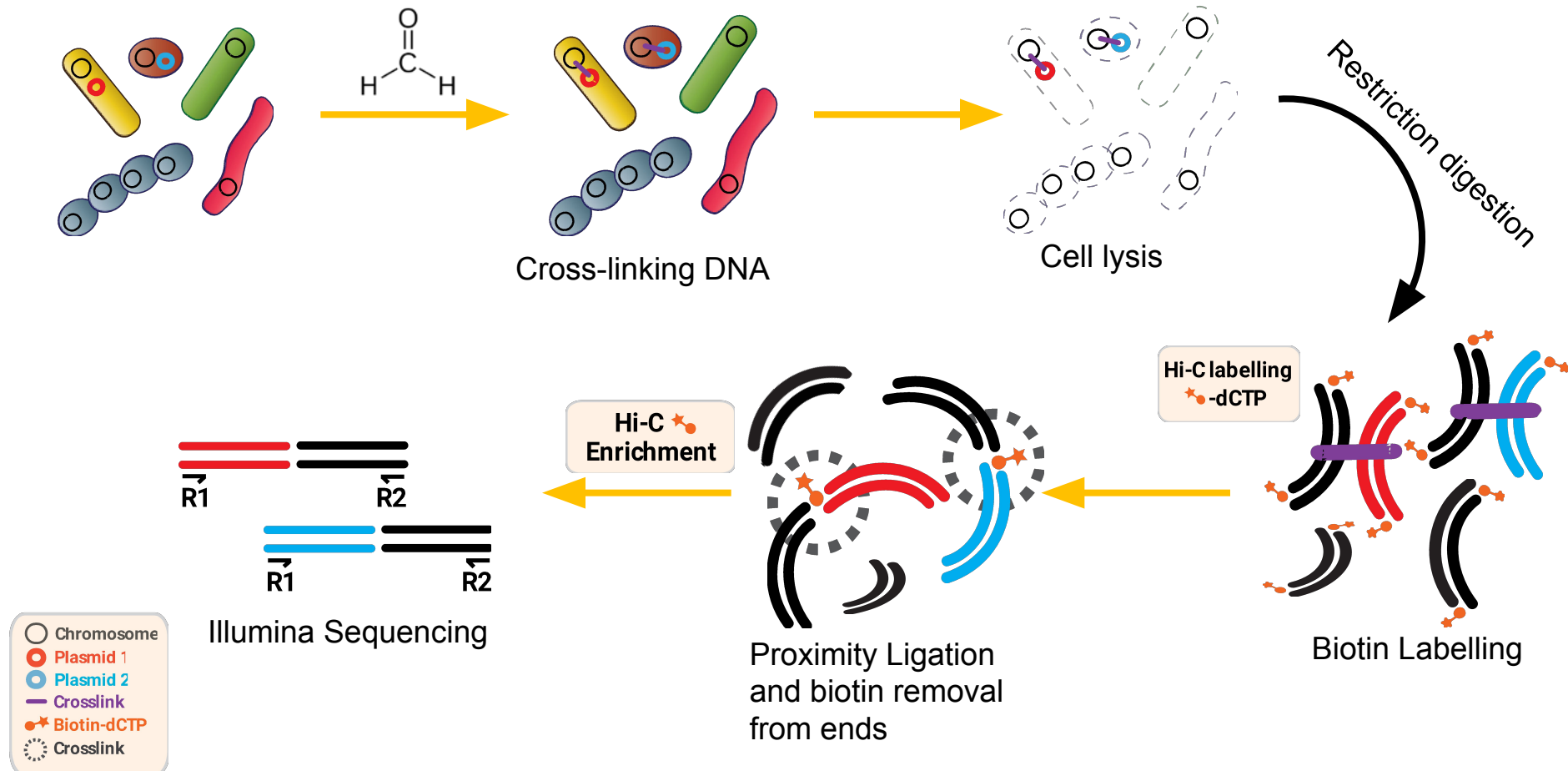
- Culture everything – culturomics
 - Money and time consuming
 - Specific approach to each environment
 - Sub-populations can be lost
- Sequence everything – metagenomics
 - No place for mobile genetic elements
 - Unreliable steps during the analysis
- Separate cells and investigate them individually

Aims and main method steps

- Describe bacterial diversity and composition with high reliability
- Detect antibiotic resistance genes / mobile genetic elements
- Linking resistance genes to bacterial species without the need of culturing







Basic steps during metagenomics sequencing analysis

Pre-processing

Quality control, raw-read filtering, merging, ...

De novo
assembly

Building up contiguous sequence fragments
from overlapping short reads

Binning

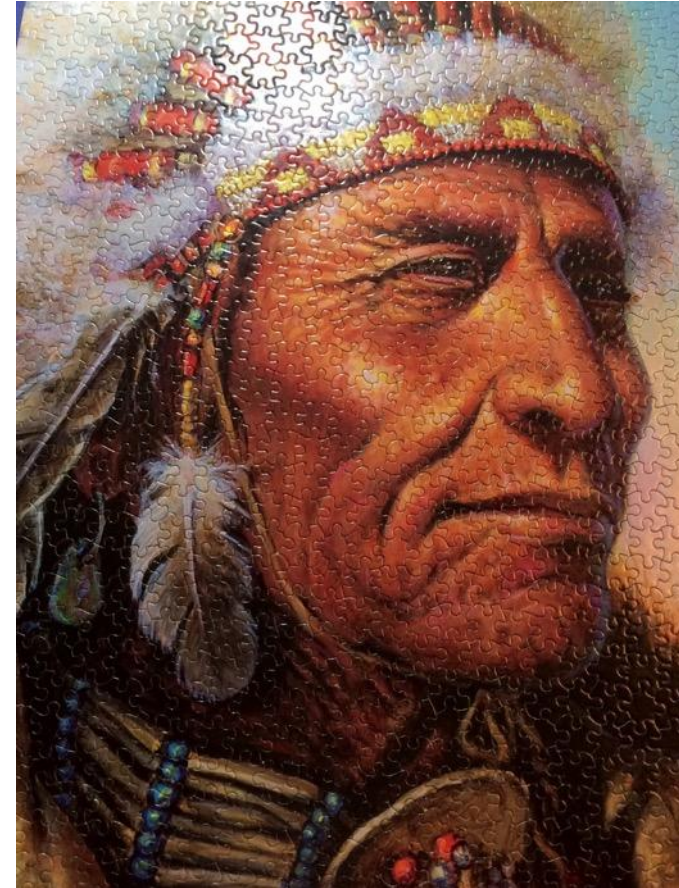
Decide which contigs belong to the same
group/cluster/assembly

Post-processing

Characterise, speciate, analyse assemblies



Binning – assembling a puzzle without knowing the final picture...



...but with a lot of pieces and high
amount of homology



Number of pieces:

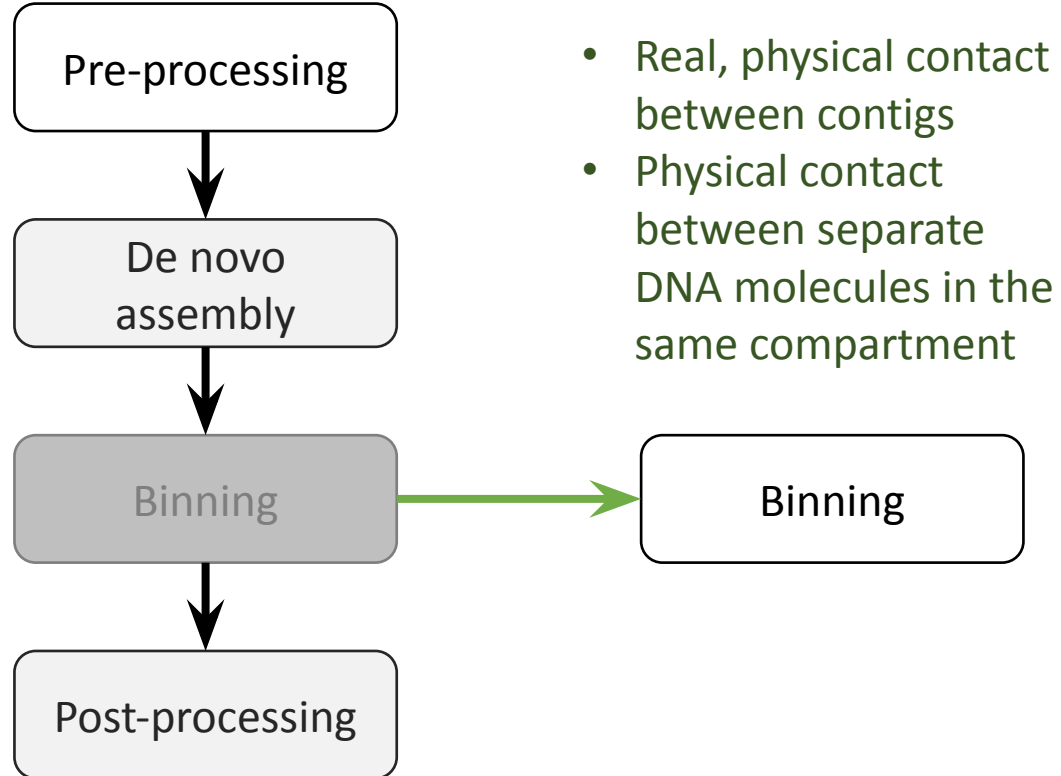
~900,000

How many dogs we have?

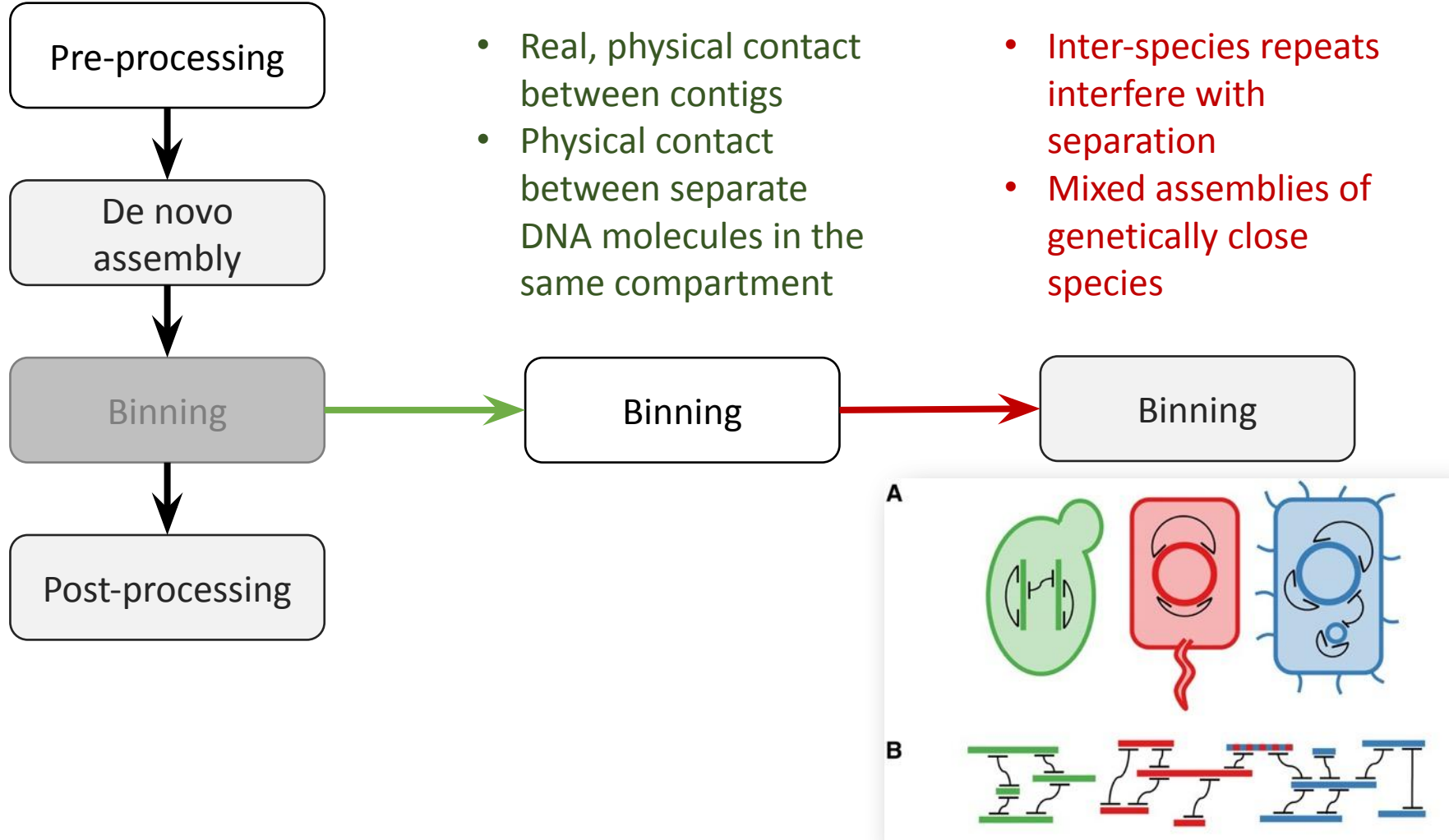
What are their names?

How many and what kind of spots
they have?

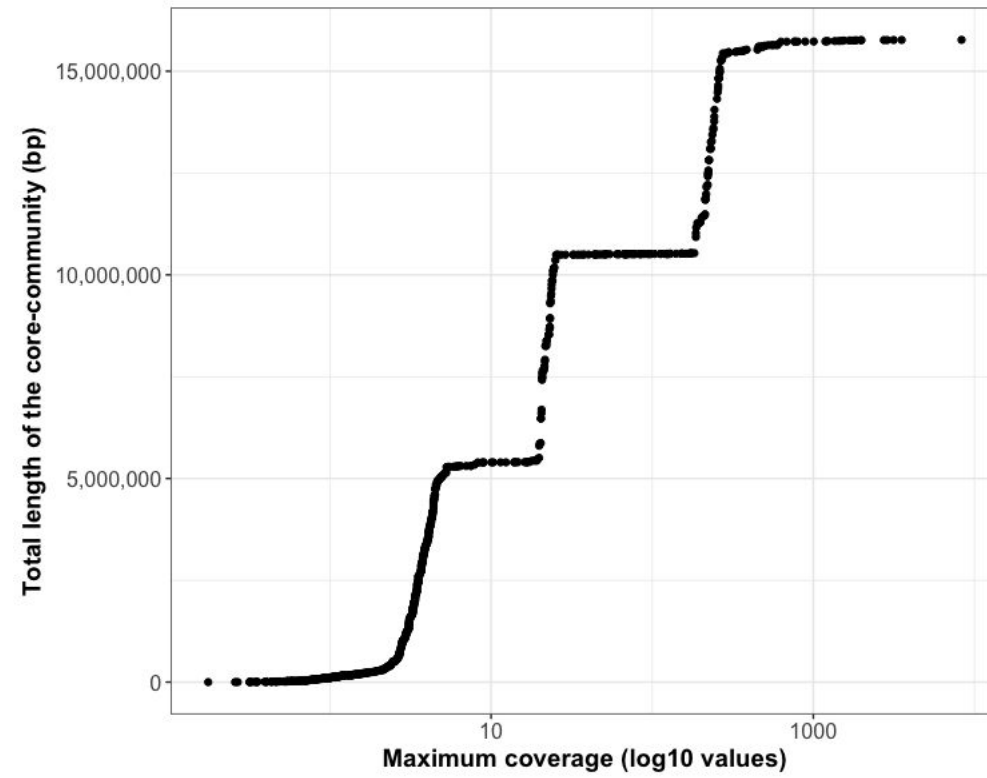
Hi-C metagenomics, advantages



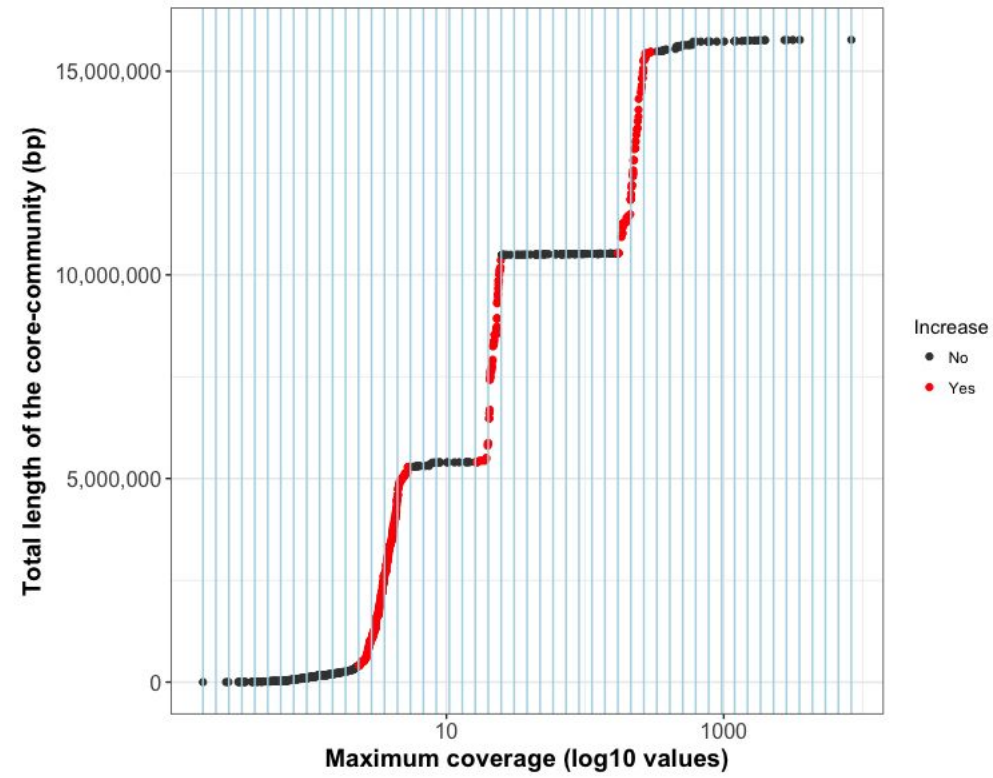
Hi-C metagenomics, disadvantages



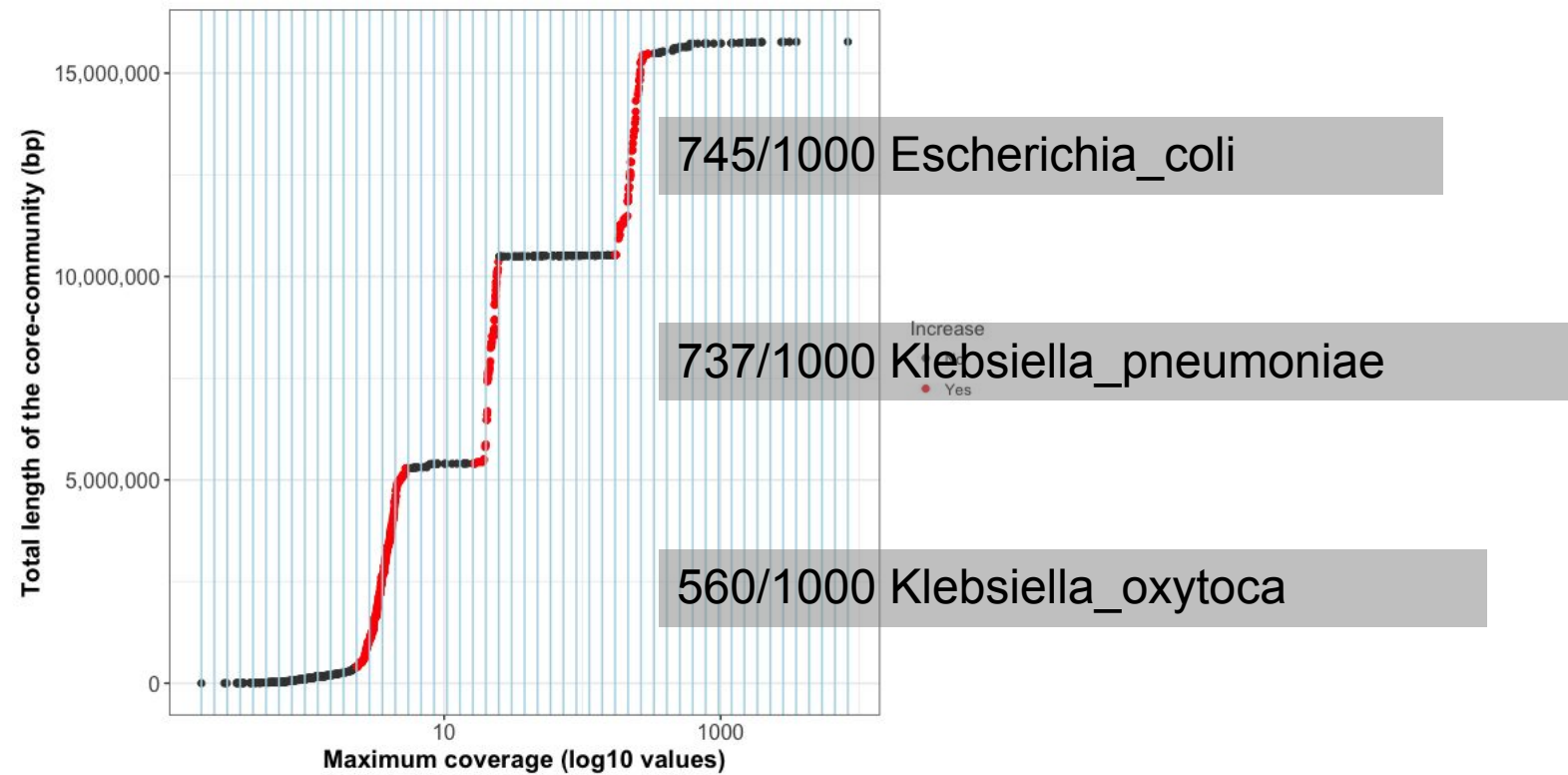
Coverage based splitting



Coverage based splitting

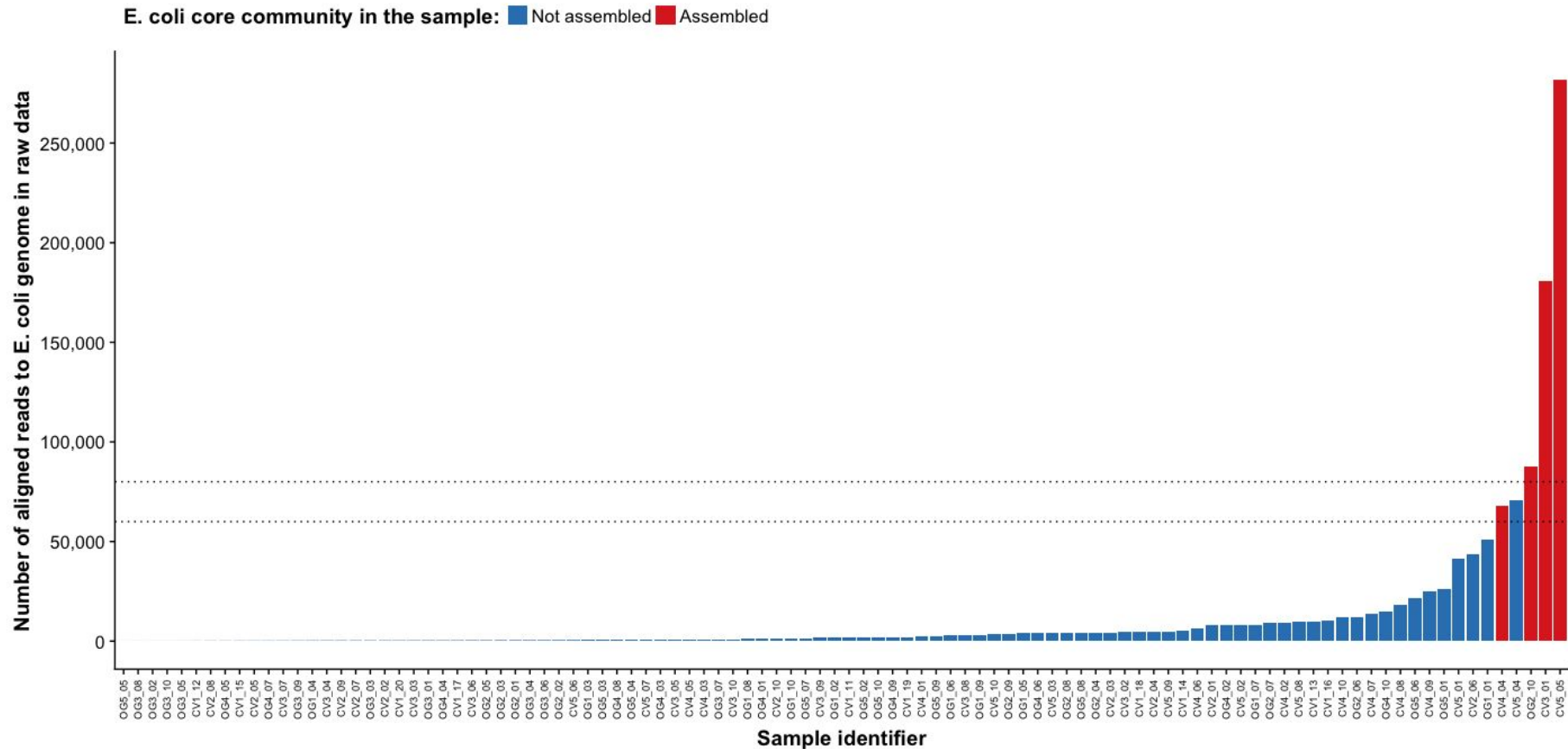


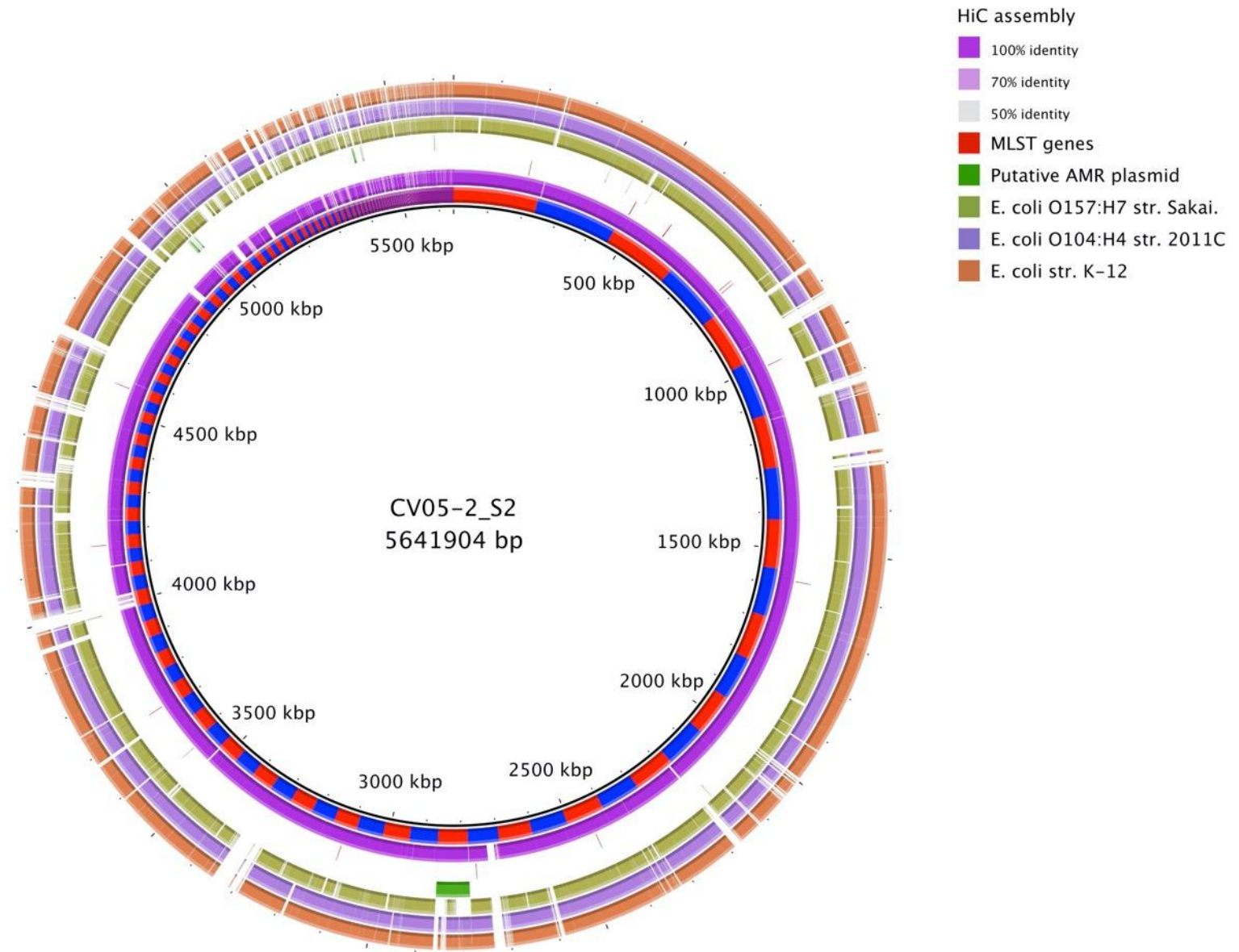
Coverage based splitting



Most of the *E. coli* levels were under our sensitivity threshold

We are assembling genomes with an abundance of 0.2% in the original community





Hi-C metagenomics summary

- Improved binning efficiency especially for less studied complex microbial communities
- Multiple steps are dealing with mixed assemblies
- High sensitivity and specificity for a dedicated set of genes (AMR genes in our case)
- Universal technique for any sample with a bacterial community in it

Hi-C metagenomics future application

- Research
 - Resistance spreading, transmission
 - Understanding plasmids in more detail
 - Mapping "exotic" microbiomes
- Monitoring
 - Individual, farm, region, country level
 - Longitudinal studies
- Diagnostics
 - Not a single day pipeline yet
 - Combined with culture-based techniques, improving antibiotics therapy